



JURONG JUNIOR COLLEGE
2017 JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

17S

EXAM INDEX

CHEMISTRY

9729/01

Paper 1 Multiple Choice

14 September 2017

1 hour

Candidates answer on separate paper.

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and exam index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** or **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

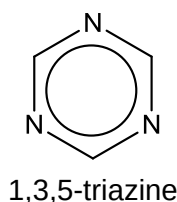
This document consists of **13** printed pages and **1** blank page.

- 1 When a beam of protons ($^1\text{H}^+$), travelling at the same speed, passes through an electric field of constant strength, the angle of deflection is $+12^\circ$.

Assuming an identical set of experimental conditions, which particle would be deflected through an angle of $+8^\circ$?

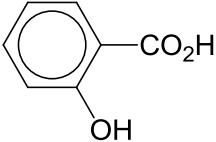
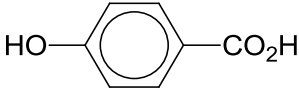
- A $^6\text{Li}^{4-}$ B $^6\text{Li}^{2+}$ C $^3\text{He}^{2+}$ D $^{12}\text{C}^{3-}$

- 2 1,3,5-triazine is an aromatic compound and its structure resembles that of benzene. The structure of 1,3,5-triazine is shown below.

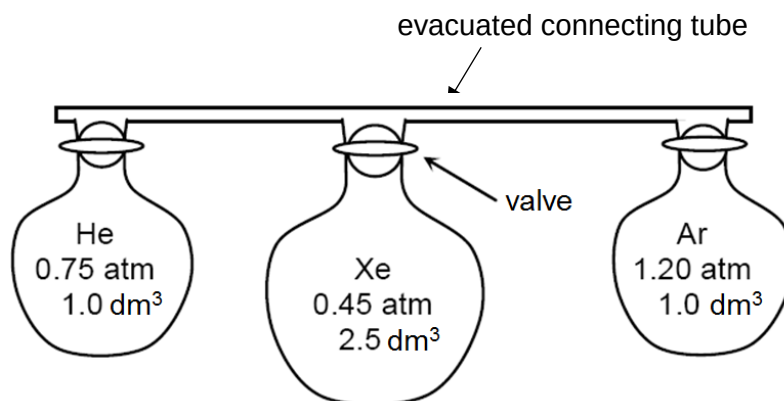


Which statement about 1,3,5-triazine is correct?

- A The empirical formula of 1,3,5-triazine is CN.
 B A molecule of 1,3,5-triazine has six σ bonds.
 C The π bond between carbon and nitrogen atoms is formed by $\text{sp}^2\text{-sp}^2$ overlap.
 D All the carbon–nitrogen bonds have equal bond lengths.
- 3 In which pair does the first substance have a higher boiling point than the second?

	First substance	Second substance
A	HF	HCl
B	CH_3COCH_3	$(\text{CH}_3)_2\text{NH}$
C	$(\text{CH}_3)_3\text{CH}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
D		

- 4 The following diagram shows the contents of three vessels which are joined together by a connecting tube which is evacuated of air.



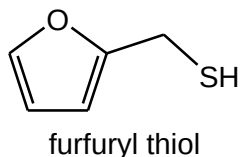
After the valves on the vessels are opened, the final pressure is measured and is found to be 0.675 atm.

Assuming that the temperature of the whole setup is maintained at 30 °C, what is the volume of the connecting tube?

- A** 0.023 dm³ **B** 0.040 dm³ **C** 0.056 dm³ **D** 0.094 dm³

- 5 *Use of the Data Booklet is relevant to this question.*

Furfuryl thiol ($M_r = 114$) is the key component of the aroma of roasted coffee.



Which statements about furfuryl thiol are correct?

- 1 On complete combustion, 0.10 g of furfuryl thiol produces 0.23 g of CO₂.
- 2 On complete combustion, 0.10 g of furfuryl thiol produces 21 cm³ of SO₂ measured under r.t.p.
- 3 On complete combustion, 0.10 g of furfuryl thiol produces 0.16 g of H₂O.
- 4 0.10 g of furfuryl thiol reacts with an excess of bromine to produce 0.38 g of product. Assume that the -SH group in furfuryl thiol does not react with bromine.

- A** 1, 2 and 4 only
B 1, 3 and 4 only
C 2 and 3 only
D 2 and 4 only

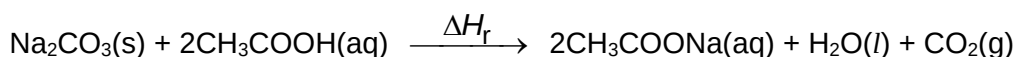
- 6 A solution mixture contains both sodium carbonate ($M_r = 106$) and sodium hydrogen carbonate ($M_r = 84.0$).

25.0 cm³ of the solution mixture was titrated against 0.300 mol dm⁻³ hydrochloric acid. 15.0 cm³ of HCl is required to decolourise the phenolphthalein indicator. A few drops of methyl orange indicator was then added and a further 20 cm³ of HCl is required for methyl orange indicator to change colour.

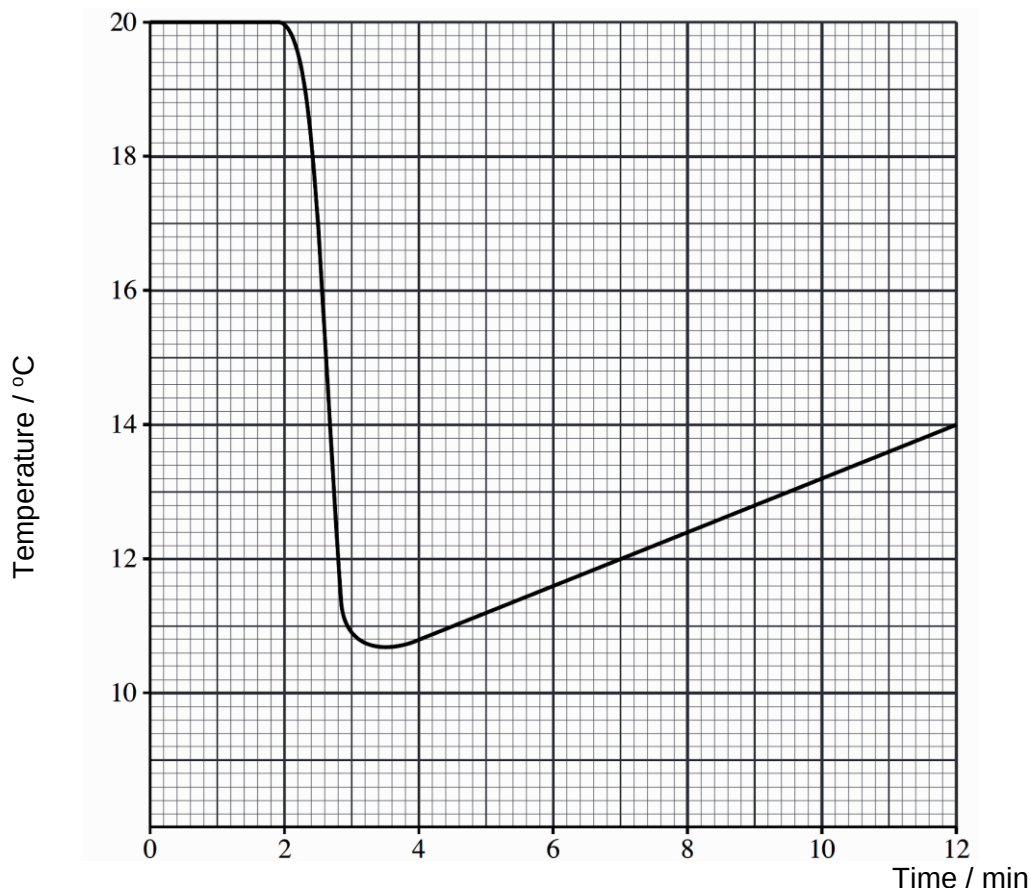
What is the percentage, by mass, of sodium carbonate present in the solution mixture?

- A 20.9 % B 25.0 % C 75.0 % D 79.1 %

- 7 Sodium carbonate reacts with ethanoic acid according to the equation below.



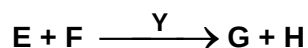
In an experiment to determine the enthalpy change of reaction, ΔH_r , 7.5 g of solid sodium carbonate ($M_r = 106$) was added to 50 cm³ of excess aqueous ethanoic acid at the second minute. The temperature of the resultant solution was monitored at various time intervals and the following graph was obtained.



Given that the specific heat capacity of the solution is 4.18 J cm⁻³ K⁻¹, what is the most accurate enthalpy change of reaction, ΔH_r , which can be obtained from this graph?

- A +17.7 kJ mol⁻¹ B +27.5 kJ mol⁻¹
C +29.5 kJ mol⁻¹ D +31.0 kJ mol⁻¹

- 8 The kinetic of the following reaction is studied, using a colorimeter, by finding the time taken for the coloured reactant, **E**, to decolourise. The reaction is catalysed by **Y**.



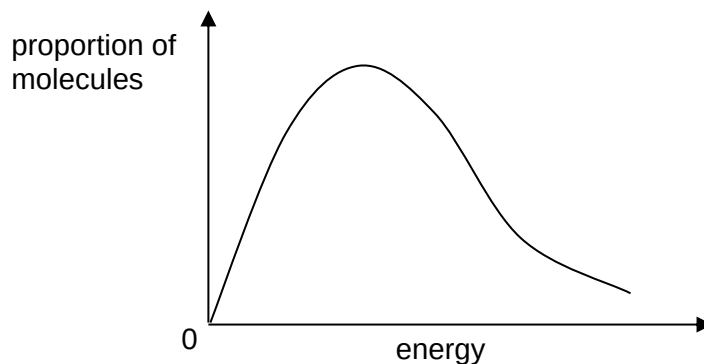
The following results are obtained:

Experiment	Volume of solution added / cm ³				Time taken / s
	E	F	Y	H ₂ O	
1	10	20	10	10	20
2	10	10	10	20	40
3	10	20	5	15	40
4	5	20	10	15	10

What is the rate equation for the reaction?

- A** rate = $k[\text{E}][\text{F}]$ **B** rate = $k[\text{E}][\text{Y}]$
C rate = $k[\text{F}][\text{Y}]$ **D** rate = $k[\text{E}][\text{F}][\text{Y}]$

- 9 The diagram represents the Boltzmann distribution of molecular energies at a given temperature.



As the temperature increases, which statements are correct?

- 1 The maximum of the curve is displaced to the right.
- 2 The proportion of molecules with energies above any given value increases.
- 3 The proportion of molecules with any given energy increases.

- A** 1 only
B 1 and 2 only
C 2 and 3 only
D 1, 2 and 3

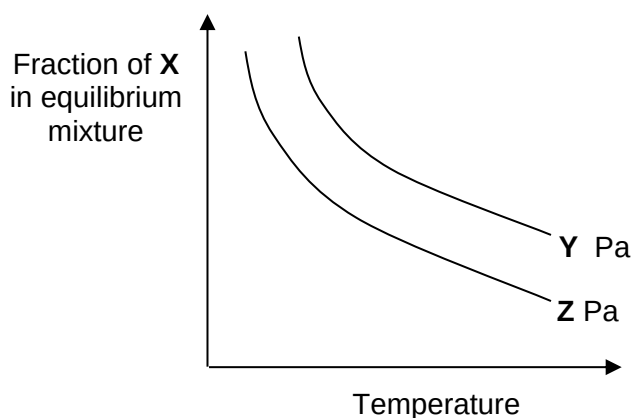
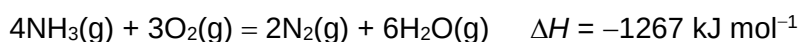
- 10 The interhalogen compound BrF_3 is a volatile liquid which autoionises.



The electrical conductivity of BrF_3 decreases with increasing temperature.

Which statement about the autoionisation process is correct?

- A The autoionisation process is endothermic and the shape of the cation is linear.
 B The autoionisation process is endothermic and the shape of the cation is non-linear.
 C The autoionisation process is exothermic and the shape of the cation is linear.
 D The autoionisation process is exothermic and the shape of the cation is non-linear.
- 11 The graph below shows how the fraction of **X**, which represents one of the following compounds in the equilibrium mixture shown below, varies with temperature at pressures of **Y** Pa and **Z** Pa.



Identify **X** and the correct relative magnitudes of **Y** and **Z**.

	X	Pressure
A	N_2	Z > Y
B	O_2	Y > Z
C	H_2O	Y > Z
D	NH_3	Z > Y

- 12 What is the pH of an aqueous solution containing 0.1 mol dm^{-3} sodium benzoate and 0.01 mol dm^{-3} benzoic acid? [K_a (benzoic acid) = $6 \times 10^{-5} \text{ mol dm}^{-3}$]

- A 3.22 B 4.22 C 4.78 D 5.22

- 13** When 25.0 cm^3 of a $0.100 \text{ mol dm}^{-3}$ strong monobasic acid was titrated against $0.200 \text{ mol dm}^{-3}$ of a weak base **Q**, 12.50 cm^3 of the base was required to completely neutralise the acid.

Which statements are correct for this experiment?

- 1 **Q** is a diprotic base.
- 2 The pH of the solution at the equivalence point is 7.
- 3 A solution of maximum buffering capacity was formed when 6.25 cm^3 of the base was added.

- A** 3 only
B 1 and 2 only
C 2 and 3 only
D None of the above

- 14** The numerical values of the solubility products of three metal sulfides are tabulated below.

Compound	Numerical value of solubility product
CuS	6.3×10^{-36}
ZnS	1.6×10^{-21}
Bi_2S_3	1.1×10^{-73}

Which of the following shows the three metal sulfides arranged in order of increasing molar solubility in water?

	lowest	→	highest
A	Bi_2S_3	CuS	ZnS
B	ZnS	CuS	Bi_2S_3
C	ZnS	Bi_2S_3	CuS
D	CuS	Bi_2S_3	ZnS

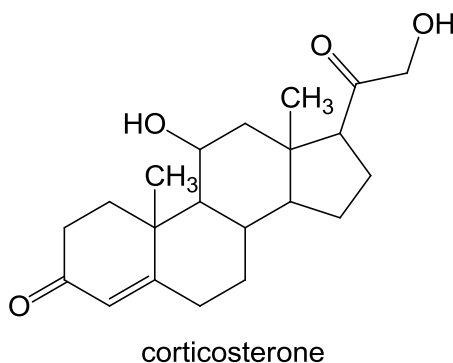
- 15** Four compounds of some Period 3 elements are listed below.



Which pair of compounds, when mixed in water, could give a resultant solution of pH 7?

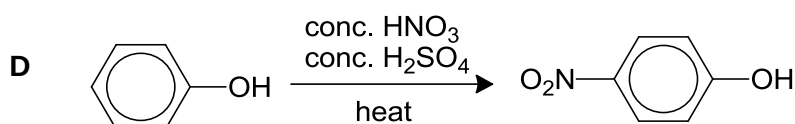
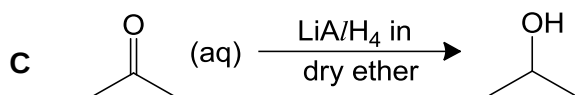
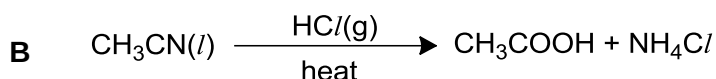
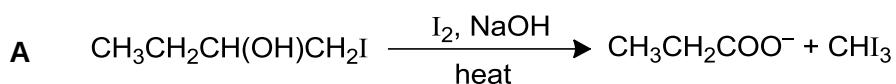
- A** NaCl and Na_2O **B** NaCl and SO_2
C Na_2O and SiCl_4 **D** SiCl_4 and SO_2

- 16** Corticosterone is a hormone involved in the metabolism of carbohydrates and proteins.



How many possible stereoisomers does Corticosterone has?

- A** 2^6 **B** 2^7 **C** 2^8 **D** 2^9
- 17** Which of the following transformation will occur under the given conditions?

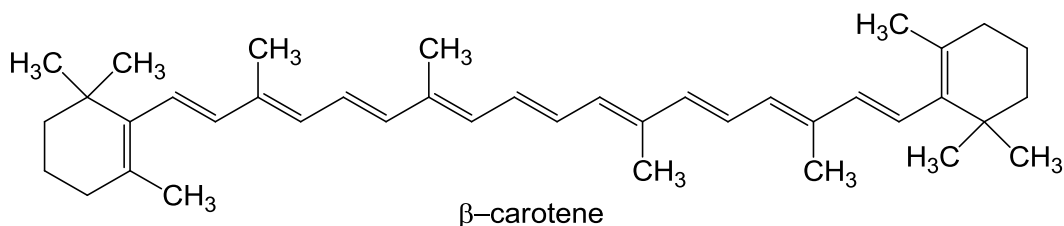


- 18** Gem-diols, in which both hydroxyl groups are bonded to the same carbon, can spontaneously eliminate a molecule of water to produce a carbonyl compound.

Which compound, after complete hydrolysis, gives a precipitate when warmed with $[\text{Ag}(\text{NH}_3)_2]^+$ complex?

- A** 1,1-dibromobutane
B 1,2-dibromobutane
C 2,2-dibromobutane
D 2,3-dibromobutane

- 19 β -carotene is responsible for the orange colour of carrots and its structure is shown below.



When a molecule of β -carotene is oxidised by hot, concentrated acidified KMnO_4 , many product molecules are formed.

How many of these product molecules contain the ketone functional group?

- A 4 B 6 C 9 D 11

- 20 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$, causes pain when its concentration builds up in muscles.

Which reagents react with **both** of the $-\text{OH}$ groups in lactic acid?

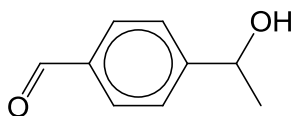
- 1 hydrogen bromide
- 2 phosphorus pentachloride
- 3 sodium
- 4 sodium hydroxide

- A 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

- 21 Which of the following represents the correct sequence of increasing $\text{p}K_{\text{a}}$ value of the four compounds?

	smallest $\text{p}K_{\text{a}}$	—————→			largest $\text{p}K_{\text{a}}$
A	$\text{CCl}_3\text{CO}_2\text{H}$	$\text{CH}_3\text{CO}_2\text{H}$	$\text{C}_6\text{H}_5\text{OH}$	$\text{C}_2\text{H}_5\text{OH}$	
B	$\text{CCl}_3\text{CO}_2\text{H}$	$\text{CH}_3\text{CO}_2\text{H}$	$\text{C}_2\text{H}_5\text{OH}$	$\text{C}_6\text{H}_5\text{OH}$	
C	$\text{C}_6\text{H}_5\text{OH}$	$\text{C}_2\text{H}_5\text{OH}$	$\text{CCl}_3\text{CO}_2\text{H}$	$\text{CH}_3\text{CO}_2\text{H}$	
D	$\text{C}_2\text{H}_5\text{OH}$	$\text{C}_6\text{H}_5\text{OH}$	$\text{CH}_3\text{CO}_2\text{H}$	$\text{CCl}_3\text{CO}_2\text{H}$	

22 Consider the following compound **G**.



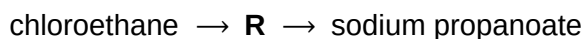
G is heated with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to form compound **H**.

Both **G** and **H** are then separately warmed with Fehling's solution and the observations are noted.

What are the observations?

- A** Both **G** and **H** give a red precipitate.
- B** Only **G** gives a red precipitate.
- C** Only **H** gives a red precipitate.
- D** Neither **G** nor **H** gives a red precipitate.

23 Chloroethane can be used to make sodium propanoate.



The intermediate, **R**, is treated with boiling aqueous sodium hydroxide to give sodium propanoate.

Which reagent would produce the intermediate, **R**, from chloroethane?

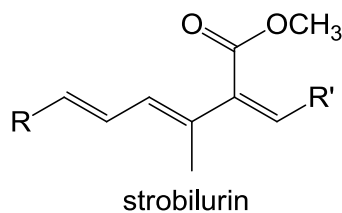
- A** alkaline KMnO_4
- B** hydrogen cyanide
- C** sodium hydroxide in ethanol
- D** potassium cyanide in ethanol

24 Which statements about the formation of a carboxylic acid are correct?

- 1 A carboxylic acid can be produced by oxidation of a primary alcohol.
- 2 A carboxylic acid can be produced by reduction of a nitrile.
- 3 A carboxylic acid can be produced by reduction of an aldehyde.
- 4 A carboxylic acid can be produced by hydrolysis of an acyl chloride.

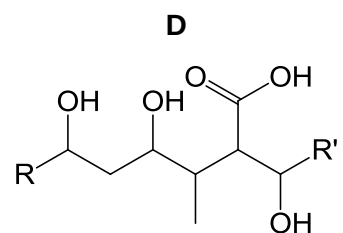
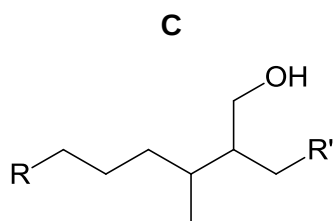
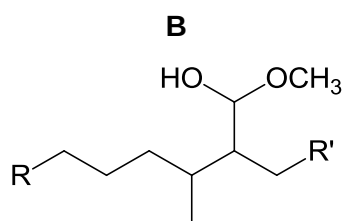
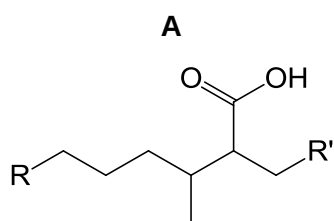
- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

- 25 Part of the structure of a fungicide, strobilurin, is shown. R and R' are inert groups.



In this reaction, strobilurin is warmed with aqueous sulfuric acid to give compound X. Compound X is then treated with hydrogen in the presence of a nickel catalyst to give compound Y.

What could be the structure of compound Y?



- 26 Why are amides, RCONH_2 , less basic than amines, RNH_2 ?

- A** Amides form a zwitterion in which the nitrogen atom carries a positive charge.
- B** Amides have a resonance structure involving the movement of a pair of electrons from the nitrogen atom to the oxygen atom.
- C** Electrons on the nitrogen atom move on to the C–N bond giving it some double bond character so that it is more difficult to break.
- D** The amide carbonyl group withdraws electrons from the NH_2 group to make the hydrogen atoms acidic.

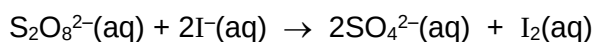
27 Which conditions are necessary when an electrode potential is measured using a standard hydrogen electrode as the reference electrode?

- 1 The use of hydrogen gas at 1 bar and 298 K.
- 2 Measurement of e.m.f. when the current delivered by the cell is effectively zero.
- 3 A pH of 1.0 for the solution at the hydrogen electrode.

- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

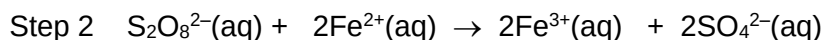
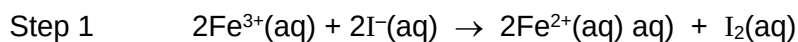
28 *The use of Data Booklet is relevant to this question.*

Peroxodisulfate ions convert iodide ions into iodine slowly.



The rate of the reaction can be increased by adding catalysts such as aqueous iron(III) ions.

A possible pathway of a catalyst reaction involves the following steps:



Which statement about the reaction is **incorrect**?

- A** $\text{S}_2\text{O}_8^{2-}$ is a stronger oxidising agent than Fe^{3+} .
B The reaction between $\text{S}_2\text{O}_8^{2-}$ and I^- is slow because the $E^\ominus_{\text{cell}}$ value is small.
C The $E^\ominus_{\text{cell}}$ for step 2 is more positive than step 1.
D Aqueous cobalt(II) ions can be used as a catalyst for this reaction.

29 The following data refers to iron as a typical transition element and to calcium as an s-block element.

For which property are the data under the correct element?

	Property	Fe	Ca
A	density	7.87	1.54
B	electrical conductivity	9.6	85
C	melting point	810	1538
D	metallic radius	0.231	0.140

- 30** When aqueous ammonia is added to a solution containing hexaaquacopper(II) ions, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, a pale blue precipitate is formed. When excess ammonia is added, blue precipitate dissolves to form a dark blue solution.

What are the roles of the ammonia molecules in the above reactions?

- 1 Bronsted–Lowry base
- 2 Ligand
- 3 Reducing agent

- A** 1 only
- B** 2 only
- C** 1 and 2 only
- D** 2 and 3 only

